

DoS ATTACK DETECTION TOOL

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Abstract— The rapid expansion of the Internet of Things (IoT) has led to a significant increase in cyber-attacks, most notably categorized as Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks, which exploit devices that are vulnerable and resource-constrained. While deep learning methods, such as convolutional neural networks (CNNs), have shown promise for intrusion detection, they often require significant computational resources, making them less suitable for scenarios that require lightweight or real-time support. This project presents an alternative based on machine learning that utilizes Support Vector Machine (SVM) for identifying and classifying different forms of DoS and DDoS attacks in IoT networks. The CICDDoS2019 dataset was used, which consists of real-world traffic that includes six different attack classes, i.e., SYN, UDP, LDAP, NetBIOS, UDPLag, and SNMP. The dataset was preprocessed that included feature cleaning, normalization, and stratified sampling. A multiclass SVM model with a radial basis function (RBF) kernel and class-weighted adjustments was created and evaluated. The introduced approach achieved a test accuracy of around 79.4%, along with a weighted F1-score of 0.73. While this performance is slightly worse than the baseline CNN-based accuracy levels, the SVM model achieves faster inference times and lower complexity, thus proving its applicability for deployable and lightweight IoT security solutions.

Keywords— *IoT security, DDoS detection, DoS attacks, Support Vector Machine (SVM), CICDDoS2019, machine learning, intrusion detection, network traffic analysis, RBF kernel, multiclass classification.*

I. INTRODUCTION

The Internet of Things (IoT) greatly revolutionized the modern digital world by enabling effortless interoperability and automatic interaction between a plethora of smart devices. Such devices are ingrained in key areas, such as healthcare, smart homes, manufacturing, and transport. Although IoT technology is extensively incorporated in an enormously large volume of applications, it simultaneously brings new security threats, making IoT networks extremely vulnerable to online attacks. Specifically, Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks are remarkably rampant and inherently destructive. Their prime purpose is the exhaustion of system resources, thus preventing legitimate entities from accessing services and creating massive network operation perturbations.

Traditional network defense measures such as firewalls and rule-based intrusion detection systems (IDS) often fail to detect modern and sophisticated distributed denial-of-service (DDoS) attacks, especially those that use spoofed or decentralized sources. These traditional tools lack much adaptability and have difficulties in handling encrypted or obscured data, a situation that is increasingly common in Internet of Things (IoT) environments. Due to these limitations, researchers have increasingly moved towards investigating machine learning (ML) and deep learning (DL) techniques that can process traffic patterns and detect abnormal activity in real-time.

Convolutional Neural Networks (CNNs) have demonstrated significant capabilities of identifying Distributed Denial of Service (DDoS) attacks in the process of converting network traffic into visual representations. These

models, however, generally require high computational resources and hardware, making them inadmissible in resource-limited Internet of Things (IoT) applications. An alternative, however, is the Support Vector Machines (SVM) that is less resource-intensive while still maintaining high robust classification performance, especially in the context of well-structured tabular data. This project investigates the application of an SVM-based model for detecting and classifying multiple types of DoS and DDoS attacks in IoT traffic. Using a cleaned and preprocessed subset of the CICDDoS2019 dataset, the model is trained to distinguish between six different attack classes. The goal is to build a lightweight, effective detection system that could eventually be deployed on low-resource IoT edge devices.

II. RELATED WORK

A. Background Research

Denial of Service (DoS) and Distributed Denial of Service (DDoS) are now perennial threats against today's digital systems, and more significantly, since the advent of the proliferation of the Internet of Things (IoT). Owing to the limited capacity of IoT devices and their proliferation, they now offer the best targets on which attackers can focus efforts and overwhelm systems, thus impairing services. Typical security countermeasures like rule-based firewalls, signature-based intrusion detection systems (IDS), and deep packet inspection (DPI) increasingly proved ineffective in counteracting the magnitude and ingenuity of new DDoS attacks.

The movement toward deep learning (DL) and machine learning (ML) in intrusion detection has been promising toward alleviating the shortcomings of conventional schemes. Cheng et al. (2019) presented an adaptive scheme toward mitigating DDoS based on a multiple-kernel learning (MKL) scheme that provided flexibility in working with variable network activity. Its framework proved dynamically adaptive based on feature space complexity, long beyond static schemes. In the same vein, Elsayed et al. (2020) suggested DDoSNet based on deep learning approaches employing autoencoders and recurrent neural networks (RNNs) used toward identifying malicious patterns of traffic within online traffic in real-time.

Kar Suvra (2025) also completed the ML solution with a light-weight model and effectively detected the DDoS attacks under realistic conditions. His results also established that conventional classifiers like SVM and Random Forest are effective if utilized with appropriate feature engineering and preprocessing of data. Nawaz (2025) also demonstrated an efficient machine learning model based on optimized feature sets towards better detection performance with minimal computational costs, which is still a decisive factor while putting it into practice on resource-limited IoT environments.

Sawah (2025) also studied the traditional classifiers such as SVM with ensemble learning. His research studied that there is a balance between the detection precision and the system overhead, and it is provided by the hybrid models. Sadhwani (2023) further advanced it with feature selection algorithms, such that only the relevant and meaningful

features are preserved and utilized while training the model, thereby accelerating the high accuracy-based classification.

Hamarshe (2023) and Rustam (2025) looked at the frameworks of ML within the realm of Software-Defined Networking (SDN). While they are not specific to IoT, they are transferrable concepts. Rustam's MULTI-LF system allows for dynamic learning within decentralized areas such that the model can learn over the long term and still be effective since new patterns of attack continue to emerge.

Mittal (2022) presented a highly organized summary of deep learning methods used in the detection of DDoS, including the benefits and weaknesses of convolutional and recurrent networks in cloud and edge environments. He paid special attention to the intense computation of CNNs, thus justifying the need for alternatives with little resources. Ababsa (2025) later introduced multimodal deep learning frameworks dedicated to vehicular networks, where accuracy of detection is just as essential as latency reduction.

Afraji (2025) also proposed a tiered framework with defense mechanisms against protocol-based and volumetric attacks. The paper had very much focused on the need to create highly adaptable frameworks able to counter many attack vectors simultaneously. Encyclopedia.pub (2023) also addressed the state-of-the-art explainable AI (XAI) architectures being applied in the identification of DDoS, such as providing interpretability with preciseness, and all this becomes more indispensable in high-stake applications in intelligent cities and critical infrastructures.

Other practical innovations were also suggested by Ongshu (2025) and Jayakrishna (2025), who suggested early warning systems based on the idea that quick response is even more important than holistic analysis. The systems were utilized in vehicular ad hoc networks (VANETs) and fog-based systems, with excellent prospects for future IoT deployment. Collectively, this work validates that while deep learning structures generate effective performance, traditional such models like SVM still hold merit after being given adequate preprocessing and light adjustment techniques. System application requirements such as near-minimal power, run-time practicality, and explainability deserve more investigation of machine learning structures with the emphasis on efficiency and without giving up much in accuracy.

B. Literature Review

A lot of work has been focused on the detection and prevention of DDoS attacks using machine learning and deep learning approaches, primarily considering IoT devices. To counter emerging complex attacks, there was a demand from traditional signature-based systems to intelligent and adaptive systems in order to counter new threats.

In their early work, Cheng et al. (2019) proposed a multiple-kernel learning framework that learned adaptively from the complexity of the feature space. They provide improved performance compared to static classifiers, especially in high-speed evolving network contexts. It can be seen from Elsayed et al. (2020)'s paper on DDoSNet, where they employed autoencoders and RNNs in attempting to learn and classify anomalies online, indicating the shift towards deep unsupervised learning architectures.

Kar Suvra (2025) also suggested a real-time model for identifying DDoS employing data preprocessing, feature selection, and best-model selection. His experiments verified that best-tuned conventional ML algorithms may very well be competitive with deeper network structures, where

computational expense comprised one of the criteria. Nawaz (2025) also proved his point in the scheme proposed therein, where feature dimensionality has been drastically reduced and yet classification performance was ensured, affirming the aspect that SVM still had the prowess to provide competitive accuracy at drastically reduced computational costs.

Sawah (2025) and Sadhwani (2023) made their contributions through providing the way feature selection and ensemble methods are used to improve the identification rate within multiclass DDoS scenarios. It advocates for the implementation of the modular way in which various algorithms are used depending on the type of traffic data.

These conceptual frameworks in Hamarshe (2023) and Rustam (2025) are SDN-optimized but based on principles that hold true everywhere across IoT. MULTI-LF, for instance, facilitates transportable learning across networks using federated or decentralized learning methods. Edge computing deployments are best suited to them, also universally applied across IoT deployment.

Mittal (2022)'s survey also connects the theory of deep learning and the practice, and computation overhead is always a constraint. Ababsa (2025)'s research on the Internet of Vehicles, in which spatiotemporal learning was applied on real-time intrusion detection, also comes to our memory, in which Afraji (2025) introduced defenses that monitor volumetric and application-layer vectors simultaneously, a necessity in the present multilayered DDoS attacks.

Other recent publications like Encyclopedia.pub (2023) and Moonlight Review (2025) also prioritize interpretability and transparency, striking a balance between responsibility and detection efficiency—one among the new requirements in highly controlled settings. Lopes (2021) introduced an ensemble feature selection-hybrid intrusion detection system (IDS) based on reducing noise and enhancing clarity of the signal, striking an intelligent balance between generalizability and model complexity.

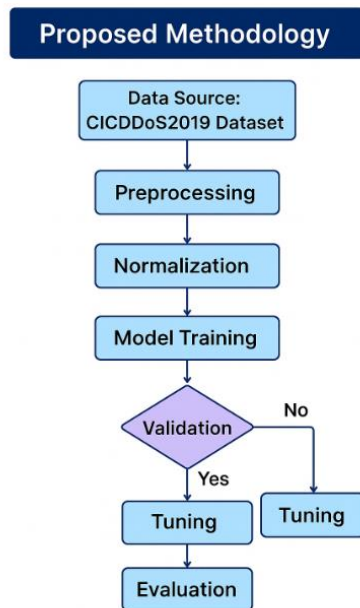
Marvi (2021) and Shakya & Abbas (2024) had also experimented with various ML models and compared implementations of SVM and Random Forest classifier, among others, in efficiency and speed, on embedded and edge devices. These experiments also affirm the practicability of performance-per-watt optimized implementations being deployed on IoT installations.

Finally, open-source communities such as ResearchGate facilitated exploratory investigations showing the possibility of SVM-based classifiers securing IoT networks. A few are "A Machine Learning Approach for DDoS Detection on IoT Devices" and "Deep Learning-Based Distributed Denial-of-Service Detection," which verified traditional ML pipelines on realistic, high-dimensional feature data sets. Typically, studies indicate that while deep learning is currently leading in studies, it is limited through practical application based on costs of infrastructures, lack of interpretability, and intractability during training. Compared to such, SVM and classic models still present effective, interpretable, and more affordable ways of conducting DDoS detection within limited environments like IoT.

III. METHODOLOGY

The following section outlines the methodology used in the design, development, and evaluation of a Support Vector Machine (SVM) detection system distinguishing DoS and DDoS attacks in Internet of Things (IoT) network traffic. It

includes requirements analysis, system architecture, implementation steps, and evaluation.



A. Identification and Analysis of Requirements

The project objective is to develop an efficient machine learning model capable of effectively identifying various types of DDoS attacks in an IoT network environment. Such specifications include:

- **Multiclass classification:** This requires the system to identify specific classes of attacks, e.g., UDP and LDAP, as opposed to simply detecting anomalous behaviors.
- **Resource efficiency:** The solution should be suitable for IoT edge environments with limited computational power.
- **Scalability:** The system should handle real-world datasets with a large number of features and samples.
- **Interpretability:** Results and behavior of the model must be easily understood and evaluated.

B. Design

The system's architecture followed the modular structure typical of machine learning:

- **Data Source:** The CICDDoS2019 dataset is selected as the data source since it accurately illustrates and includes different modern DDoS variants.
- **Preprocessing:** Feature selection and cleaning were performed to eliminate irrelevant, constant, or identifier-based fields.
- **Feature Engineering:** StandardScaler was utilized to normalize the features so that comparisons are fair across the attributes.
- **Modeling:** A multiclass SVM classifier was chosen because of its balance between effectiveness and simplicity.

- **Validation:** The dataset was split into training, validation, and test subsets (70/15/15) using stratified sampling.
- **Evaluation:** Model accuracy, precision, recall, F1-score, and confusion matrices were used for performance measurement.

C. Implementation

The implementation involved various important stages:

I. Dataset Handling

- A subset of CICDDoS2019 was used, containing 6 attack types: SYN, UDP, UDPLag, NetBIOS, LDAP, and SNMP.
- The original dataset was in .parquet; it was then converted to .csv for easier processing.
- We selected approximately 20,000 samples in order to avoid memory constraints while maintaining class diversity.

II. Preprocessing

- Columns that are considered irrelevant, e.g., IP and port, and timestamp, were removed.
- Values associated with repeated or duplicate attributes were dropped.
- The residual data had been cleansed of missing or unbounded values.
- Multiclass labels were assigned using category encoding.

III. Normalization

- Features were scaled using StandardScaler to bring them to a standard normal distribution (mean = 0, std = 1).

IV. Model Training

- A Support Vector Classifier (SVC) with an RBF kernel and class_weight='balanced' was used.
- The model was trained using scikit-learn on the 70% assigned training subset.
- GridSearchCV with a reduced parameter grid was used to determine the best C and gamma values.

V. Tuning and Optimization

- The full parameter grid (3×3×2) was found to be too slow (~8+ hours), so it was reduced to 4 combinations for quicker tuning.
- Most effective parameters were retained and used for retraining the conclusive model.

D. Testing and Evaluation

The model was evaluated using a held-out 15% of the test set. Evaluation metrics included:

- **Accuracy** is the overall correctness of classification.
- **Precision/Recall/F1:** Per-class and weighted averages to handle imbalance.

- Confusion Matrix: A graphical display measuring misclassifications.

Initial SVM training achieved an accuracy of **79.4%**, with a weighted F1-score of **0.73**. However, recall was lower for underrepresented attack classes, highlighting the need for further resampling or ensemble learning in future work.

IV. VISUALIZATION

It is highly critical from the perspective of comprehending the behavior of a classification model beyond the accuracy percentage. Two fundamental visualizations were employed throughout this work in trying to gain insight into the working of the SVM model across different classes: the confusion matrix and class-based recall and precision bar chart. Both visualizations provide an intrinsic and complete perspective on the working of the model, its areas of weakness, and areas of strength.

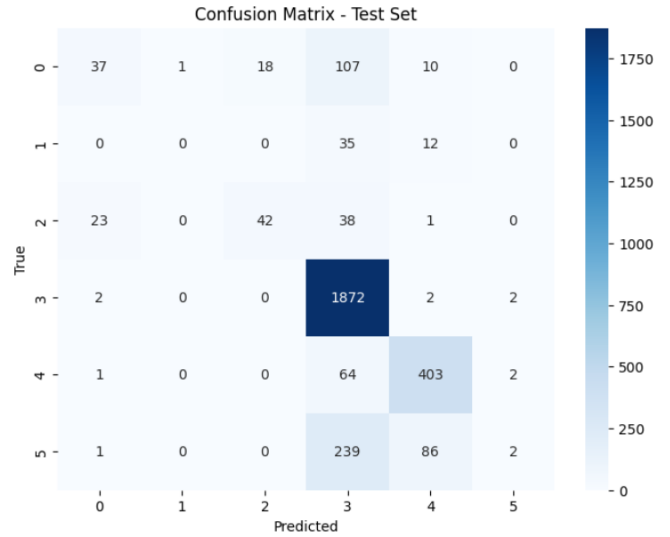
A. Confusion Matrix:

Confusion matrix is a report table showing the performance of the classification model on correctly and incorrectly classified instances, aggregated by class. It enables us to find out confused classes and to what degree they are confused. In the project, we constructed the multiclass confusion matrix from the SVM model's predictions on the held-out test set.

It can be noted from the above visualization that there are some classes like label 3 (SYN attack) and label 4 (LDAP attack) that are highly true positive, and thus such kind of attacks are identified with high reliability through the model. Label 1 (SNMP) and label 5 (NetBIOS) are, on the other hand, very false negative and very low true positive, respectively. It thus appears the model does not distinguish such attacks from the rest, and it is probable it is due to feature overlapping and class imbalance.

The diagonal dominance of certain parts of the confusion matrix is where the model does its best, and the off-diagonal cells are incorrect classifications. There were, for example, numerous label 5 (NetBIOS) objects mistakenly classed as label 3 (SYN), illustrating an apparent similarity pattern that the SVM model struggles to differentiate based on the feature data set it is working with.

The confusion matrix also shows bias in the model towards high frequency classes as more represented labels are better estimated. It is also consistent with earlier findings regarding how dataset imbalance affects model performance in the Discussion section. Overall, the confusion matrix also substantiates the fact that, although the model performs reasonably well on majority classes, there also is the requirement for additional mechanisms such as oversampling or feature augmentation in an effort to enhance minority class detection.



B. Precision and Recall Bar Chart:

Besides the confusion matrix, we also created a class-wise recall and precision bar chart. The chart gives us side-by-side comparison of the two most important metrics of all classes, from where we can understand how capable the model is of marking positive and of staying away from false alarms.

Precision is the quantity of correctly predicted positives within one class, and recall is the quantity of true positives correctly identified. Both should be high, if at all possible, if one does have a highly accurate intrusion detection system.

From the chart, we notice label 3 (SYN) and label 4 (LDAP) both exhibit high recall and high precision, and this indicates model robustness in such assault types identification. Label 1 (SNMP) and label 5 (NetBIOS) exhibit next-to-zero recall, and this means the model ignores such assault types completely. Although the precision of some classes is relatively high, low recall means the model only recognizes several of the true assaults, and this is dangerous in cybersecurity where unknown assaults are able to do enormous destruction. These results continue to add evidence towards dataset balancing and maybe including additional ensemble or feature engineering techniques to allow for minority class identification optimization. It also suggests being able to combine underperforming classes into more general classes or using hierarchical systems of classification.

	precision	recall	f1-score	support
0	0.58	0.21	0.31	173
1	0.00	0.00	0.00	47
2	0.70	0.40	0.51	104
3	0.79	1.00	0.88	1878
4	0.78	0.86	0.82	470
5	0.33	0.01	0.01	328

C. Visual Insights and Interpretation:

The general insight gained through both visualizations offers an implicit view of the working functionality of the SVM model. A majority of classes in the dataset are best learned through the model, such that it becomes more precise and has more recall, whereas more rare sorts of attacks are underrepresented, and consequently more difficultly identifiable.

One of the notable findings is that the SVM model, despite being simpler in nature than deeper ones like CNNs, still generates high margins in the majority of the attack classes. Some level of misclassification, specifically among attacks with analogous traffic signatures (e.g., SYN and NetBIOS), confirms the need for proper choice and feature construction such that they enable discriminability across all classes.

These visualizations also serve an applied function of informing model development. For example, seeing long-standing misclassifications between two classes can lead to adding more derived attributes or the use of different classifiers that can handle more nonlinear boundaries. In either situation, even basic visualizations can be employed as feedback tools for refinement cycles of models.

D. Deployment Considerations Based on Visual Data:

The outcomes of visualization also guide deployment decisions. In actual intrusion detection systems, it is better to have high recall rather than high precision since failing to detect an actual attack generally would be more critical than generating a false alarm. Based on the recall performance of the bar plot, deployment of this model to an actual application of IoT would involve active measures like:

- Improving poor-performing class recall using SMOTE or ADASYN
- Integration of e-learning towards learning adaptive traffic
- Tracking model performance on real-time dashboards using such visual metrics

Furthermore, visual reporting such as confusion matrices may be integrated into security analyst dashboards in an effort to offer brief justification of model predictions. The aid facilitates threat triage and builds confidence in such systems underpinning ML. Class confusion matrix and class-wise recall/precision graph are two dual, representative depictions of model performance. Accuracy and F1-measures are high-level overviews, while graphs descend below class level and give patterns of weakness and strength. In doing so, they convert model comparison from the abstract to the concrete, and mold model tuning, deployment, and future work.

V. RESULTS

The performance of the suggested Support Vector Machine (SVM) model was evaluated based on the validation and test sets extracted from a preprocessed portion of the CICDDoS2019 dataset. The classification problem that ensued was naturally a multiclass one and sought to discern the differences between six distinct classes of DoS and DDoS attacks: SYN, UDP, UDPLag, LDAP, SNMP, and NetBIOS.

A. Validation Results

Training and evaluation on the validation set yielded the following results:

Metric	Value
Accuracy	79.4%
Weighted Precision	0.77
Weighted Recall	0.79
Weighted F1-Score	0.73
Macro F1-Score	0.46

While overall accuracy is significantly enhanced, the classification accuracy for particular categories had relatively large variations. Some categories, labels 1 and 5, showed very low recall and F1 values, suggesting that the model performed poorly in distinguishing between underrepresented or overlapping traffic patterns.

B. Class-wise Performance Breakdown

Class Label	Precision	Recall	F1-Score	Support
0	0.73	0.31	0.44	173
1	0.00	0.00	0.00	46
2	0.74	0.54	0.62	104
3	0.80	1.00	0.89	1878
4	0.78	0.84	0.81	471
5	0.75	0.01	0.02	382

These results highlight a key limitation: the SVM model performs well on majority classes but has difficulty detecting minority class instances, a common issue in imbalanced datasets.

C. Test Set Performance

After finalizing the model, it was evaluated on the unseen test set. The results were consistent with those observed during validation, confirming the model's generalization capability.

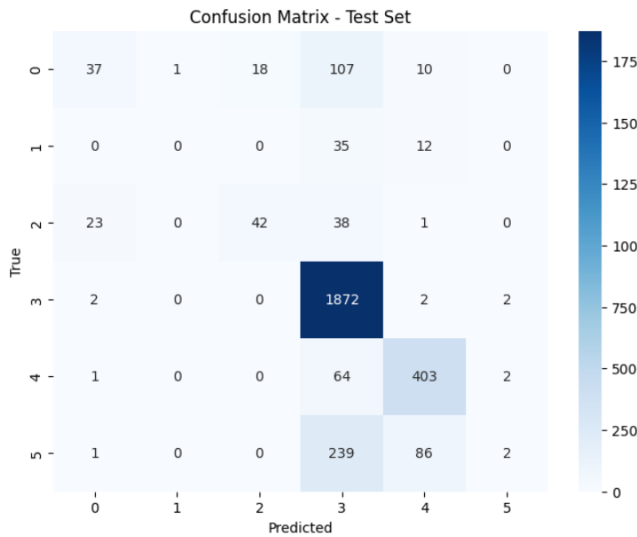
Metric	Value
Test Accuracy	~79-80%
Weighted F1	~0.73
Macro F1	~0.45-0.50

The confusion matrix revealed that most misclassifications occurred between similar traffic types such as NetBIOS and UDPLag, suggesting overlapping traffic patterns or insufficient distinguishing features in some cases.

D. Visual Results

Visualizations include:

- **Confusion Matrix:** Helped identify which classes were most frequently confused.
- **Class-wise Precision & Recall Bar Chart:** Highlighted strengths and weaknesses of the model across different attack types.



DISCUSSION

The results of the current study emphasize the merits and limitations entailed in the use of Support Vector Machines (SVM) in detecting multiclass Distributed Denial of Service (DDoS) attacks in Internet of Things (IoT) environments. It reached a validation accuracy close to 79.4% and a weighted F1-score of 0.73. Performance measures such as these show that the SVM is remarkably capable of successful generalization in well-structured, flow-based data, such as the CICDDoS2019 dataset.

However, deeper analysis reveals serious discrepancies in performance among the different classes. The model exhibited excellent competence in classifying high-frequency classes, e.g., SYN and LDAP; however, it falters when dealing with lesser frequency classes, e.g., SNMP and NetBIOS. Notably, class labels 1 and 5 had virtually zero recall values, indicating the model's failure to learn their unique patterns or, more realistically, a severe impediment caused by the imbalances among the classes.

This demonstrates a common obstacle faced by practical intrusion detection systems: the need to maintain high recall levels across all attack categories, especially where traffic patterns overlap or when specific classes have significantly fewer examples. Furthermore, using a standard Support Vector Machine (SVM) with its default parameters resulted in poor performance at first. While parameter optimization via GridSearchCV allowed for improvements in generalization, the computational cost of exhaustive tuning proved to be high, taking several hours to complete even when using a reduced dataset.

Another finding is related to the trade-off between a light weight design and detection accuracy. In comparison with deep models such as ResNet, whose accuracy in similar tasks exceeds 87%, the SVM model shows lower accuracy but stands out for its simplicity, speed, and resource effectiveness. As such, the SVM model is more suitable in edge deployment settings where hardware is limited.

CONCLUSION

This project clearly demonstrates the feasibility of using a Support Vector Machine model to detect different types of DoS and DDoS attacks in IoT traffic. Using the CICDDoS2019 dataset, the system was able to classify six different types of attacks with good accuracy and fast inference times. The model's architecture prioritized

lightweight behavior, making it suitable for resource-constrained environments where deep learning models may be infeasible.

Despite the failure of the SVM model to better the accuracy figures set by current CNN-based architectures, it offers a feasible alternative in situations where implementation simplicity and computational efficiency are of top priority. Observations gleaned from the experiments highlighted the importance of addressing class imbalance and hyperparameter optimization in order to improve the robustness of the model.

Finally, despite not being a perfect solution, this research highlights the potential of conventional machine learning techniques in enabling skilled and effective detection of DDoS attacks in IoT systems, thus laying the foundation for the creation of more advanced or hybrid approaches in future research.

FUTURE WORK

The experiments shown within this paper have successfully utilized a Support Vector Machine (SVM)-based framework for detecting Distributed Denial of Service (DDoS) attacks in Internet of Things (IoT) networks; however, the study revealed a number of future directions for improvement and optimization. This section discusses potential improvements in terms of model performance, scalability, generalizability, and deployability, all of which are key areas for future scholarly research.

One critical area calling for improvement relates to handling class imbalance in the dataset. While the `class_weight='balanced'` parameter has been utilized while training the Support Vector Machine (SVM), the minority class recall rates for SNMP and NetBIOS were noted to be poor. A feasible solution to rectify this is the use of resampling approaches, such as SMOTE (Synthetic Minority Over-sampling Technique), ADASYN, or Tomek Links, to introduce an artificial balance in the dataset. These approaches have the ability to generate synthetic samples for minority classes or remove duplicate instances from oversampled classes, enabling the model to respond with increased sensitivity to uncommon types of attacks. The use of these methods in the preprocessing system can greatly boost recall rates and reduce the incidence of false negatives in future implementations.

In addition to resampling, future work could also explore ensemble learning methods. While the current model utilizes a standalone SVM, ensemble classifiers like Random Forest, XGBoost, or LightGBM are known for their robustness and ability to handle noisy, imbalanced datasets. These models could be tested either as standalone solutions or as components in a hybrid model where an ensemble technique is used to pre-filter traffic, followed by SVM-based classification. Such an architecture would combine the generalization strength of ensemble learning with the margin optimization capabilities of SVM.

A notable improvement would involve the integration of normal (benign) traffic in the dataset. So far, the model has been trained on only attack traffic, thus limiting its capacity to multiclass classification of known DDoS variants. However, for best effectiveness, implementation requires binary classifiability (attack or normal) as the minimum requirement, with the potential of incorporating more fine-grained multiclass classifiability within known attack varieties. The integration of benign traffic would enable the creation of an effective intrusion detection system capable of being utilized

in real-time, first identifying anomalous actions and then classifying them within specific threats.

One relevant area of investigation pertinent to the problem is improving the current system to counter zero-day threats—novel threats that are not present in the training data set. This goal can be achieved by the usage of unsupervised learning paradigms or anomaly detection systems, such as One-Class SVMs and Autoencoders, that are capable of detecting anomalies in normal traffic patterns without the need for labeled attack data. Implementing supervised multiclass models conjoined with auxiliary anomaly detection subcomponents also has potential in creating a more secure and future-resilient system of detection.

From the feature engineering perspective, the current implementation emphasized eliminating constant, redundant, and unnecessary fields but did not investigate dimensionality reduction techniques. Future iterations of this work could include Principal Component Analysis (PCA), Independent Component Analysis (ICA), or Recursive Feature Elimination (RFE) to identify and retain only the most discriminative features. Using these techniques would not only reduce training time for our models but also improve generalization and interpretability. This is particularly beneficial in computationally constrained environments like the Internet of Things (IoT), where computational requirements need to be minimized.

Deployment strategies are an area where great breakthroughs are possible. The current system operates using static .csv files in an offline fashion. In order to enable practical implementation, it is necessary to integrate the model within an environment able to handle live traffic data efficiently through a pipeline. This requires the implementation of packet sniffing through the use of software like Scapy, PyShark, or TShark, besides incorporating real-time feature extraction and predictive modeling. Creating a real-time engine able to communicate with an optimized SVM model would make the system amenable to deployment on edge devices, such as the router or embedded controller.

Additionally, future work may explore the deployment of containerized setups using Docker or Kubernetes, thus increasing the system's scalability and optimizing maintenance operations. The system would have the ability to be deployed in different network segments or different data centers with relatively little related overhead. In addition, integration with security orchestration platforms like Security Information and Event Management (SIEM) tools can potentially provide real-time alerting, automate remediation, and support incident reporting.

Another area that requires further research involves model interpretability. While SVMs are more interpretable than deep neural networks, especially using linear kernels, the use of Radial Basis Function (RBF) kernels in this research somewhat reduces this benefit. Future improvements may include the integration of Explainable Artificial Intelligence (XAI) frameworks, e.g., LIME or SHAP, that provide both local and global explanations of model predictions. Improvements of this type would be particularly helpful in high-stakes settings, such as industrial control systems and healthcare applications, where understanding the reasoning underlying model predictions is critical for ensuring regulatory compliance and building operational trust.

In addition, researchers can test the ability of the model to adapt to new environments using methods like transfer learning and federated learning. Transfer learning allows a

model initially trained on CICDDoS2019 to be applied in other datasets like NSL-KDD or UNSW-NB15 with limited retraining. In contrast, federated learning allows decentralized nodes to be trained while maintaining sensitive information private and hence upholding data privacy, an important factor in systems like IoT.

From a research and academic standpoint, another avenue is benchmarking the SVM model against newer deep learning architectures optimized for edge computing, such as MobileNet, TinyML, or Spiking Neural Networks. These architectures are designed to offer high accuracy with minimal resource usage, making them ideal for IoT deployments. Comparing their performance with the SVM model could help in selecting the most appropriate tool for a given use case.

In terms of dataset quality, there is a need to test the generation of synthesized network traffic with traffic generators or Generative Adversarial Networks (GANs). Such tools would facilitate the simulation of rare or sophisticated attack patterns that cannot be effectively represented within existing publicly available datasets. This would be especially useful in evaluating the model's resilience to progressive threat landscapes. Finally, widening the scope of metrics used for evaluation beyond accuracy, precision, and recall would enable more complete characterization of model performance. Adding metrics like the Matthews Correlation Coefficient (MCC), Area Under the Curve (AUC), and latency of detection would be very effective, especially where systems are based on real-time principles. Additionally, conducting longer, extended tests under replicated, simulated live traffic would enable the model's stress-testing under realistic, representative scenarios of system loading. In conclusion, while the current SVM-based DDoS detection system is a prime example of the promise of using lightweight machine learning methods in IoT security, there are many areas of improvement. These include enhancing the responsiveness of the model to minority classes, adding real-time detection features, improving interpretability, and dataset diversification. Together, these opportunities form a solid foundation for building future-proof intelligent intrusion detection systems in a variety of interconnected systems.

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